

CVIS Hands-on Training on Development of Machine Learning and Deep Learning Models Summer 2025

Facilitator
Sanjida Tasnim
Adjunct Lecturer



Compute Vision and Intelligent Systems Lab
Dept. Of Computer Science and Engineering
School of Data and Sciences, BRAC University

Overview:

- Data Pre-processing
- Why Preprocessing is Important
- Convolution
- Pooling and unpooling
- Flatten
- Fully Connected Layer
- Application: Digit classification
- Adversarial Example
- Network architectures



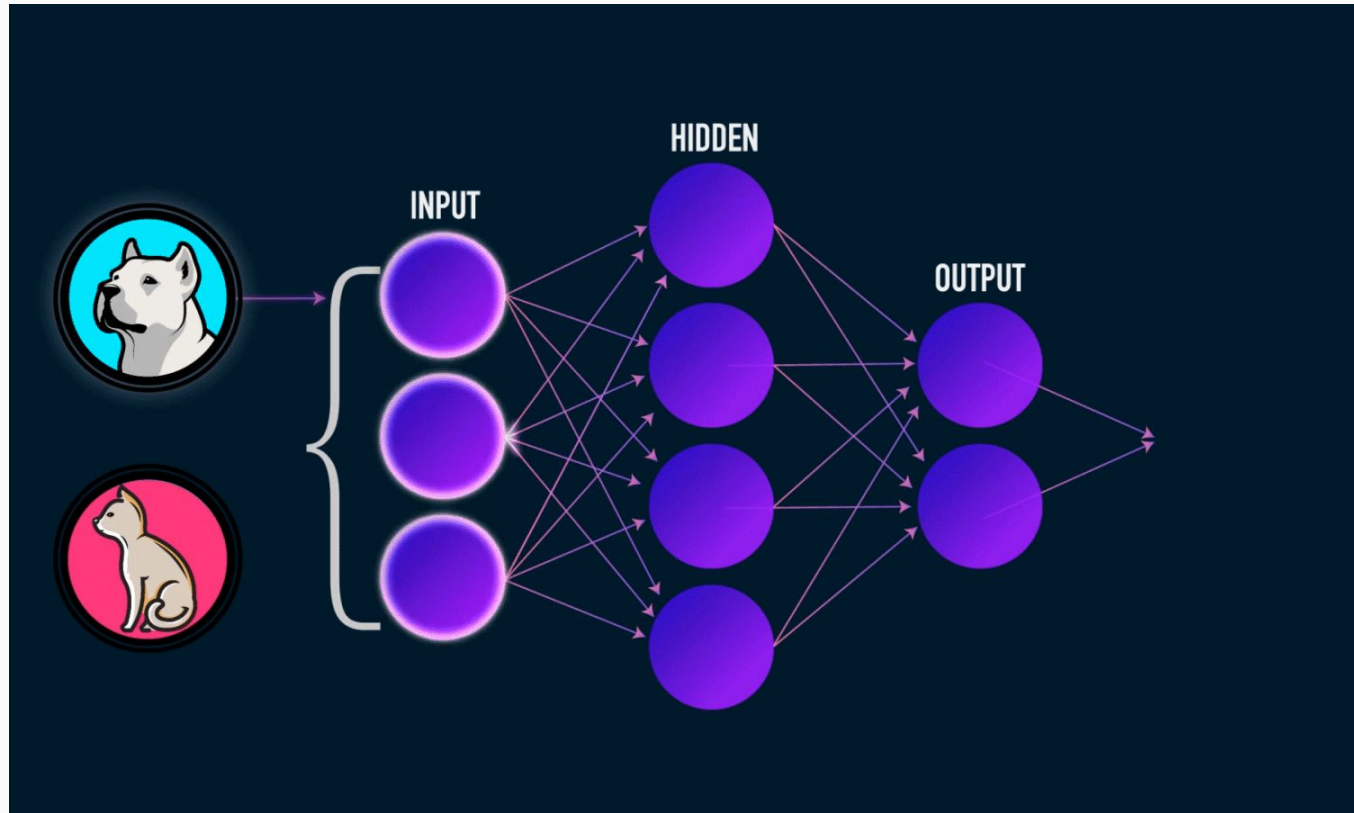
01. Understanding Image Data

- What is Image Data?
 - Represented as pixel grids (2D arrays)
 - Grayscale: One channel (0–255 intensity)
 - RGB: Three channels (Red, Green, Blue)
- Example: MNIST digit → 28×28 grayscale = 784 pixels
- Challenges:
 - Varying resolutions
 - Different lighting, noise, and orientations
 - Large size & redundancy
- https://colab.research.google.com/drive/1l6_VK9pfWZ89o1B6q609q1OoGrTHKPTA
- https://www.youtube.com/watch?v=2sUO8iE-jZs&list=PL7da2kG_V7RIIf_HXQdmMzwj4bQVEeSuh



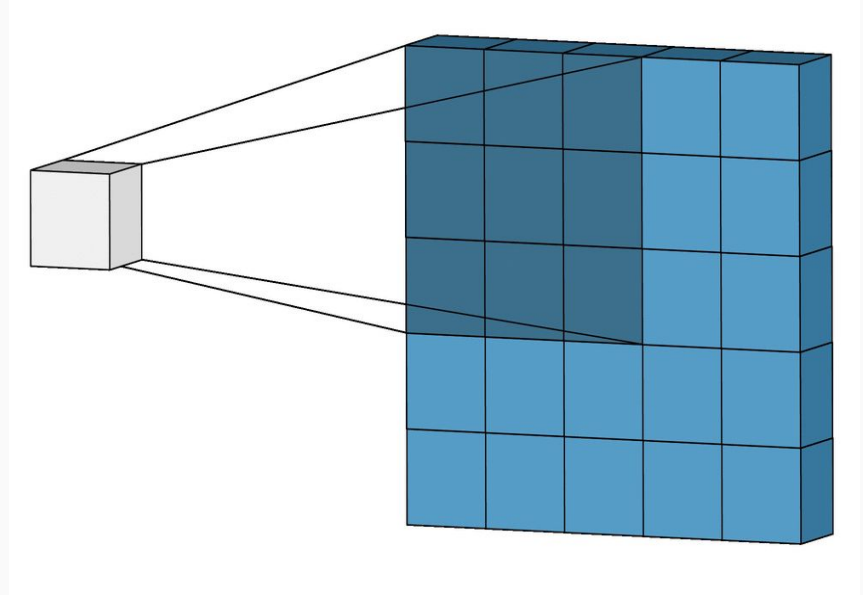
02. Why Preprocessing is Important

- Goal:
 - Reduce noise & irrelevant variability
 - Normalize scale & intensity
 - Improve convergence during training
- Benefits:
 - Faster learning (reduced training time)
 - Better generalization to new images
 - Lower risk of overfitting
- Typical Steps:
 - Resizing to a fixed input size
 - Normalizing pixel values
 - Data augmentation for robustness



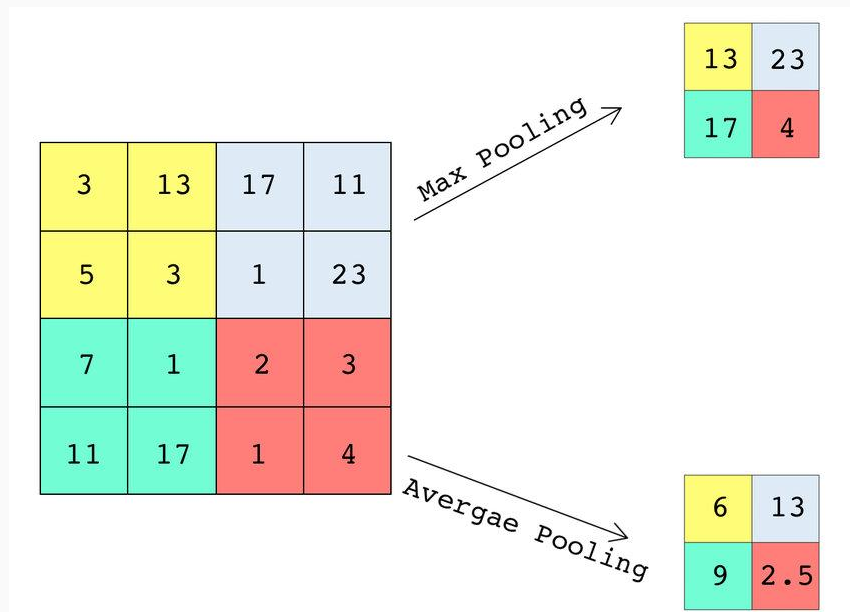
03. Convolution

- Convolution is used to extract local features using filters/kernels
- Operation:
 - Slide kernel over image
 - Multiply & sum → feature map
- Benefits:
 - Parameter sharing (fewer weights)
 - Spatial locality
- Example Kernels: Edge detection, blur, sharpen



04. Pooling & Unpooling

- Pooling: Downsampling to reduce spatial size & computation
 - Max Pooling: Keeps strongest feature
 - Average Pooling: Keeps average value
- Unpooling, aka: Upsampling
 - Reverse pooling → restores spatial resolution
 - Used in segmentation/autoencoders
- Effect: Makes features translation-invariant



05. Flatten

1	1	0
4	2	1
0	2	1

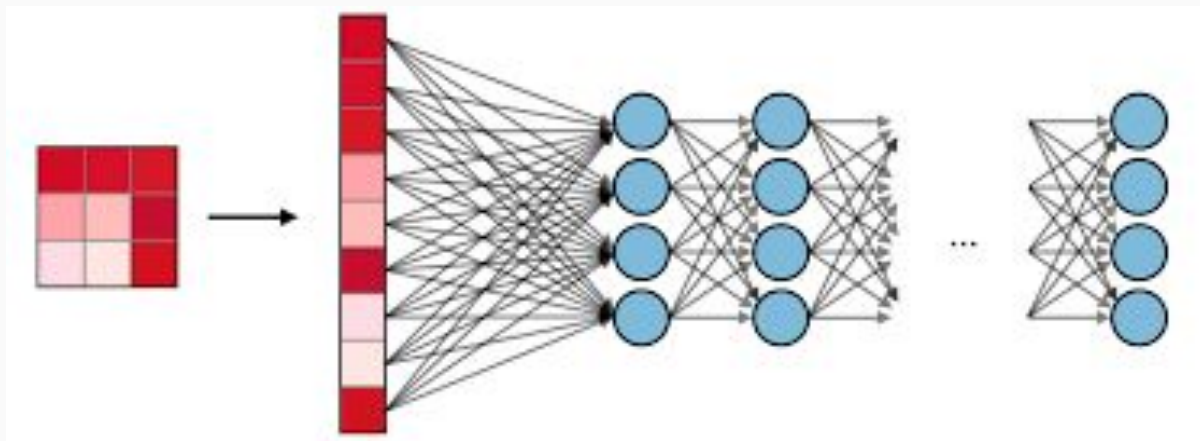
Pooled Feature Map

Flattening

1
1
0
4
2
1
0
2
1

06. Fully Connected Layer

- In CNNs fully connected layers often follow convolutional and pooling layers used to interpret the feature maps generated by these layers into the final output categories or predictions.
- In fully connected feedforward networks these layers are the main building blocks that directly process the input data into outputs.





07. Application: Digit classification

https://colab.research.google.com/drive/1ZQuqAArYh7D1G8h3sVxi9g_2BsbBGbsd?usp=sharing



08. Adversarial Examples

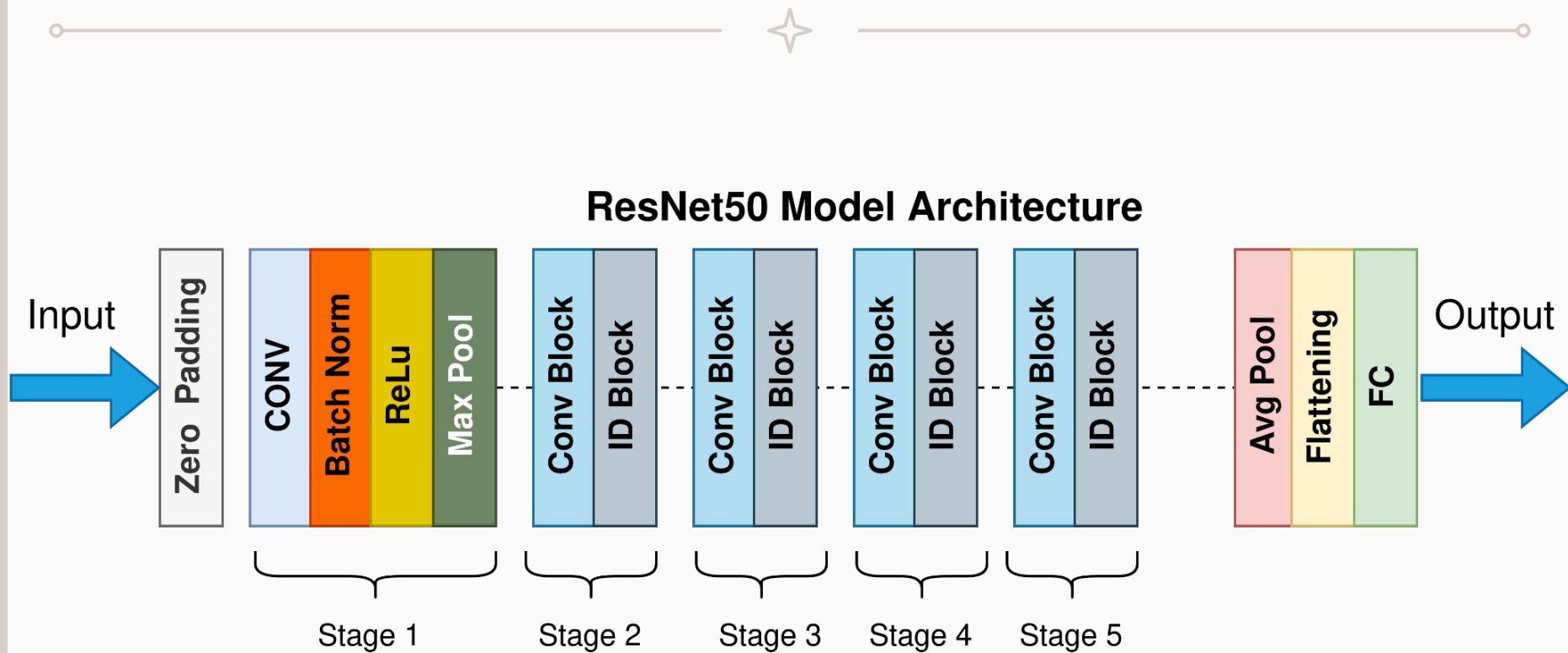
- Inputs altered slightly to cause a model to misclassify.
- Eg 1. A tiny noise added to an image of a “7” in MNIST might cause a CNN to wrongly predict “1”
- Eg 2. Unknown stop signs can cause self-driving cars to misinterpret them
- Why Do Adversarial Examples Matter?
 - Reveal security risks in AI systems.
 - Help in designing more robust models.
 - Used in adversarial training to improve resilience.



09. Network architectures

VGG - 16





<https://medium.com/@nitishkundu1993/exploring-resnet50-an-in-depth-look-at-the-model-architecture-and-code-implementation-d8d8fa67e46f>



Helpful Youtube Links

Here is a doc file containing all the links I share with my CSE 463 students-

- https://docs.google.com/spreadsheets/d/e/2PACX-1vSa1z9ODwH3s74PZo7QIPGn8kPfy0cyf75Su98ZL54sqYiU8X9DsII_I6pT8SUhHiJ6FZ8TgsBQG8H_/pubhtml?gid=1740133678&single=true